

Computational Identification and Audit of Third-MUB Loci Within Karlsson’s Three-Parameter Family

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Abstract

I document a reproducible per- H computational pipeline for mutually unbiased basis (MUB) extendability inside Karlsson’s three-parameter family $K_6^{(3)}$ only, together with a computational identification of third-MUB loci in audited search sets. Within a certified search CSV (928 rows, 899 pool-complete) and HP re-verification, the only reproducing third-MUB loci found are the F_6 $\theta=0$ bounded λ -arc and the Dita full λ -circle (Finding 1)—an identification of the audited search, not a theorem on all of $K_6^{(3)}$. Supporting HP probes (Findings 2, 3, 4), the $\mathcal{S}^*$ primary sweep (1863/1865 candidates, no fourth MUB observed), a documented 512-point coarse-grid miss of every known third-MUB locus, pool-completeness tables, and reproduction pins are included so the report is independently citable as methodology. Certified fourth-MUB obstruction at seven Dita points and exact D_0 re-verification appear in the companion theorems paper [1]; confirmatory negatives here do not replace that certificate.

Index Terms

Complex Hadamard matrices, Karlsson family, mutually unbiased bases, homotopy continuation, reproducible computation, computational audit

I. INTRODUCTION

Mutually unbiased bases (MUBs) in $\mathbb{C}^d$ satisfy

$$|\langle \psi | \phi \rangle|^2 = \frac{1}{d} \quad (1)$$

for vectors from distinct bases [4], [5]. In dimension six, Zauner conjectured $N(6) = 3$ [2], [3]. Complex Hadamard matrices (CHMs) supply second bases; Karlsson’s family $K_6(\theta, \phi, \lambda)$ [7] is a three-parameter source of 6×6 CHMs treated in the composite-dimension MUB review of McNulty–Weigert [8]. Grassl [6] showed that searches must be *per* Hadamard matrix, not via a fixed pool unbiased only to F_6 .

This companion report is a stand-alone computational methodology and audit document. Its primary contributions are: (i) a corrected per- H homotopy/cliue pipeline with formal search sets; (ii) computational identification of two reproducing third-MUB loci in audited data; (iii) honest negative evidence on larger sets, including a self-correcting 512-point grid miss that warns against over-reading sweep negatives. Algebraic gauge structure and the certified/exact fourth-MUB theorems are proved in the companion theorems paper [1].

A. Contributions

- M1** Reproducible per- H MU-pool pipeline, completeness gate, planted-cliue validation, and formal sets $\mathcal{S}_{588}$, $\mathcal{S}^*$, $\mathcal{R}$.
- M2** Computational identification of third-MUB loci in the certified search (Finding 1; F_6 $\theta=0$ arc and Dita λ -circle), with HP support (Findings 2, 3, 4).
- M3** Audit trail: 512-point coarse-grid miss; $\mathcal{S}^*$ sweep; pool-completeness tables; reproduction appendix.

II. SCOPE AND LOGICAL STATUS OF RESULTS

Scoped to $K_6^{(3)}$ and the search sets in `docs/SEARCH_SETS.md`.

- **HP-verified findings.** Locus identification and sampled Dita persistence (Findings 1, 4, 2, 3); not theorems on all of $K_6^{(3)}$.
- **Confirmatory numerics.** $\mathcal{S}^*$ and CSV `found_fourth=false`; weaker than the seven-point certificate in [1].
- **Open.** Absence of a fourth MUB on all of $\mathcal{R}$ (Open Problem 1).
- **Companion theorems [1].** T1 (gauge), T3 (`certify()` at seven points), T4 (exact GB at one D_0 -equivalent pair), and the Karlsson/ A -matrix literature correction.

III. BACKGROUND

Let H be a 6×6 CHM. Normalized columns form a basis mutually unbiased to the computational basis. Karlsson’s family is written as in [7], [8]; the companion theorems paper records the gauge algebra used below [1]. I use Karlsson’s original A -matrix formulas throughout (the McNulty–Weigert transcription fails numerical CHM tests; see [1]).

IV. METHODS

A. Problem Formulation

For fixed H , seek unit vectors $v \in \mathbb{C}^6$ unbiased to $\mathcal{B}_I$ and $\mathcal{B}_H$. With gauge $v_0 = 1$ and variables $z_i = v_i \sqrt{6}$ ($i = 1, \dots, 5$), computational unbiasedness yields $|z_i| = 1$, encoded as $z_i w_i = 1$ with auxiliary variables $w_i \approx \bar{z}_i$ on the conjugate locus.

Unbiasedness to column k of H requires

$$\left| \sum_{j=0}^5 \overline{H_{j,k}} z_j \right|^2 = 6 \iff \left(\sum_j \overline{H_{j,k}} z_j \right) \left(\sum_j H_{j,k} w_j \right) = 6. \quad (2)$$

I impose (2) for $k = 0, \dots, 4$; column $k = 5$ is dependent by Parseval once $H/\sqrt{6}$ is unitary and $w_i = \bar{z}_i$. The system has 10 equations in 10 unknowns and *depends on H* .

Critical convention: (2) uses *column* index k ($H_{j,k}$), not row index.

B. Root Finding

I use HomotopyContinuation.jl [11] with fresh polyhedral solves (mixed volume 252) and parameter homotopy from F_6 , with fresh-solve fallback when tracking drops solutions.

C. Four-MUB Extension Test

Given pool $\mathcal{P}$:

- 1) Orthogonality graph on $\mathcal{P}$ (edge if $|\langle v, w \rangle| < \varepsilon_{\text{ortho}}$).
- 2) `found_third` $\Leftrightarrow$ a clique of size ≥ 6 exists.
- 3) For each 6-clique B_3 , filter for a second 6-clique $\Rightarrow$ `found_fourth`.

D. Validation Protocol

Before sweeping, I require:

- V1** At $H = F_6$: 156 raw roots, 48 deduplicated pool vectors [9], [10].
- V2** Homotopy-with-fallback restores full pool when tracking drops roots.
- V3** At generic $K_6(0.3, 0.5, 0.2)$: pool size 48, `max_clique` = 2.
- V4** At Dita ($\arccos(1/\sqrt{3}), \pi/4, 0.4$): pool $\in \{48, 72, 120\}$ [5].

V. IMPLEMENTATION

The pipeline is implemented in Julia 1.12.6 (HomotopyContinuation 2.22, Graphs 1.14) with Python helpers:

- `src/mub_zauner_6d_liang_chen.jl` — core solver, extension test, pool completeness;
- `validate_perH_pipeline.jl`, `audit_clique_pipeline.jl` — validation and planted-clique audits;
- `sweep_karlsson.jl`, `search_special_loci.jl` — coarse and certified parameter sweeps;
- `locus_classification.jl`, `validate_third_mub_candidates.jl` — Track D HP identification;
- `dita_lambda_fourth_dense.jl`, `fourth_mub_locus_sweep.jl` — Track A/C fourth-MUB probes;
- `karlsson_k6_3.py`, `chm_equivalence.py`, `finish_project_audit.py`.

Reproduction commands and search-set definitions are in `REPRODUCE.md` and `docs/SEARCH_SETS.md`.

VI. RESULTS

A. Validation

All checks V1–V4 passed (`validate_perH_pipeline.jl`). At F_6 the fresh solve returned 156 raw solutions and 48 pool vectors after conjugate-locus filtering and deduplication.

B. Coarse Parameter Sweep (Superseded)

I initially swept $\theta, \phi \in [0.02, \pi - 0.02]$ and $\lambda \in [0, 2\pi]$ on an $8 \times 8 \times 8$ grid (512 points). Table I summarizes that run; **it must not be read as exhaustive**. The grid avoided $\theta = 0$ and missed the Dita point $(\theta, \phi, \lambda) \approx (0.955, \pi/4, 0.4)$. `found_fourth=false` on this grid is therefore not evidence of absence: the same run also reported `found_third=false` at every point, which was false.

TABLE I
COARSE PER- H SWEEP (512 POINTS, $\theta, \phi \geq 0.02$). SUPERSEDED BY RE-AUDIT IN TABLE II.

Quantity	Value
Grid points	512
found_third on grid	0
found_fourth on grid	0
max_clique on grid	2 (all sampled)

TABLE II
RE-AUDIT AFTER PLANTED-CLIQUE VALIDATION AND ANCHOR TESTING.

Point	$ \mathcal{P} $	max_clique	found_third	found_fourth
Planted orthonormal basis (ONB, synthetic)	6+decoys	6	true	—
F_6	48	6	true	false
$\theta = 0$ subfamily	48	6	true	false
Dita	72	6	true	false
Generic $K_6(0.3, 0.5, 0.2)$	48	2	false	false
Coarse-grid samples	48–72	2	false	false

C. Re-Audit: Planted Cliques, Anchors, and Certification

Table II reports post-audit results including special loci omitted from the coarse grid.

At F_6 , HomotopyContinuation certify yields **156 distinct certified** roots (mixed volume 252); pool_completeness_report gives pool_complete_flag=true with 48 deduplicated pool vectors. A full re-certification of all 512 coarse-grid points found 512/512 satisfy the corrected completeness gate ($n_{\text{certified}} = n_{\text{tracked}}$); none had $\text{max_clique} \geq 3$. Certification validates the MU-pool polynomial system, not clique number or fourth-MUB existence.

The sole Möbius degeneracy candidate from degeneracy_scan.py coincides with the Dita anchor; rounded 10-digit coordinates yield $\text{max_clique} = 2$ while exact Dita parameters yield $\text{max_clique} = 6$. Witness-set and NID probes at F_6 and Dita confirm the *per-H* MU-pool system is zero-dimensional (156 and ~ 240 certified isolated roots; Finding 5). A 20×20 grid at fixed Dita θ found 0/400 clique-6 hits because grid points omit exact $\phi = \pi/4$; targeted ϕ -offset sweeps (Table V) give clique-6 only at $\phi = \pi/4$ and drop to clique-2 for $|\Delta\phi| \geq 10^{-3}$ at all nine values. At $F_6(\theta=0, \lambda=0.3)$, clique-6 along λ drops at $|\Delta\lambda|=0.0550448$ (+) and 0.0625984 (−) (arc width ≈ 0.118). Detailed probe statistics appear in Appendix D (Tables IV–V).

D. Certified Search Set $\mathcal{S}_{588}$ and Extended CSV

I define the **certified search set** $\mathcal{S} \subset K_6^{(3)}$ as points with pool_complete_flag=true and $n_{\text{certified}} = n_{\text{tracked}}$. The design target $\mathcal{S}_{588}$ is documented in docs/SEARCH_SETS.md. The on-disk CSV results/special_loci_search.csv contains 928 certified rows (899 pool-complete, 0 with found_fourth=true) as of 2026-08-05. Reproducible third-MUB anchors: F_6 at $\theta = 0$ and Dita; $\theta = \pi/2$ circulant_match rows are **tolerance artifacts** ($\text{max_clique} = 6$ at ortho_tol = 10^{-8} but 2 at 10^{-12}). $\mathcal{S}^*$ **phased batches**: authoritative full primary sweep 1863/1865 distinct points, found_fourth=false throughout (special_loci_degen1845.meta.txt).

E. Main Computational Finding: Computational Identification of Third-MUB Loci

Track D (contribution M2)—computational identification of the audited search, not a theorem. I catalogued HP-verified clique-6 rows via locus_classification.jl. Of 45 sweep candidates with $\text{max_clique} \geq 6$, **36** survive 256-bit clique re-verification; **9** $\theta = \pi/2$ circulant_match rows fail HP. Clustering by locus proximity yields **two reproducing components** only:

- **F6_theta0 arc:** $\theta = 0$, bounded λ -interval (width ≈ 0.118); ϕ is a gauge coordinate [1] (27 HP rows).
- **Dita λ -circle:** $\theta = \arccos(1/\sqrt{3})$, $\phi = \pi/4$, clique-6 on the full λ circle; λ is a matrix-level moduli direction with π -periodic CHM class [1] (9 HP rows).

CHM-equivalence between cluster representatives gives minimum residual ≈ 5.24 (F6_theta0 vs. Dita), confirming **distinct** components. Full JSON output: results/locus_classification.json. No additional HP-verified third-MUB components appeared in the certified search CSV (928 rows) or the $\mathcal{S}^*$ primary sweep (1863 candidate points). This is an identification of the *audited* search, not a proof that no other loci exist in $K_6^{(3)}$.

Finding 1 (Third-MUB loci in certified search). *Within the certified search CSV (928 rows), HP re-verification of clique-6 candidates, and Track D clustering, the only reproducing third-MUB loci found are, up to parametrization gauge:*

TABLE III
POOL COMPLETENESS AT ANCHOR POINTS (POOL_COMPLETENESS_REPORT). DEDUPLICATED POOL SIZE $|P|$ VARIES WITH λ ON THE DITA SLICE (72–120 IN COMPANION T3 RUNS).

Point	MV	n_{cert}	$n_{\text{dist.}}$	$ P $	OK?
F_6	252	156	156	48	yes
Dita ($\lambda=0.4$)	252	240	240	72	yes
$\theta=0$ ($\lambda=0.3$)	252	156	156	48	yes
Generic $K_6(0.3, 0.5, 0.2)$	252	83	83	48	partial [†]

[†]Homotopy drop suspected; fresh solve used as fallback (drop_detection_stress_test.txt).

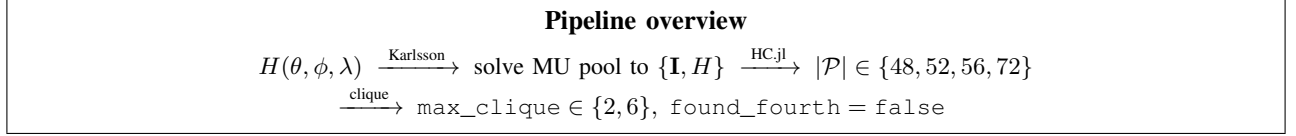


Fig. 1. End-to-end pipeline applied at each Karlsson parameter triple. HC.jl denotes HomotopyContinuation.jl.

- 1) the F_6 $\theta=0$ bounded λ -arc (width ≈ 0.118 rad), and
- 2) the Dita full λ -circle at $\phi = \pi/4$ (λ moduli direction; π -periodic CHM class).

Nine $\theta = \pi/2$ circulant_match CSV rows are non-reproducing under HP.

F. Fourth-MUB Negative Evidence (Confirmatory; Track C, Track A)

Beyond the pointwise certified obstruction of the companion theorems paper [1], this subsection records confirmatory negative numerics on larger audited sets; Open Problem 1 remains open. I define a covering region for future exhaustive search:

$$\mathcal{R} = \mathcal{S}^* \cup \{\text{Dita } \lambda\text{-circle}\} \cup \{F_6 \theta = 0 \text{ } \lambda\text{-arc}\} \cup \{\text{anchors}\},$$

where $\mathcal{S}^*$ is the full-degeneracy primary queue from search_special_loci.jl; see docs/SEARCH_SETS.md. **Status:** certified search CSV: 928 rows, found_fourth = false throughout. $\mathcal{S}^*$ primary: 1863/1865 distinct candidate points (1852 pool-complete, 21 candidate loci with max_clique = 6, 0 fourth). **Track A:** dense Dita λ probes (628/628 clique ≥ 6 , fourth = 0). **Track C:** $\theta=0$ slice scans (36 points; 6/36 third, 0/36 fourth); per-basis fourth tests on 45 clique-6 rows (270 basis tests, all negative).

Conjecture 1 (Open problem: fourth-MUB absence on $\mathcal{R}$). *For all $(\theta, \phi, \lambda) \in \mathcal{R}$ with pool_complete_flag=true, no fourth MUB exists extending $\{\mathbf{I}, H(\theta, \phi, \lambda)\}$.*

Status: Open Problem 1 unproven; the companion paper [1] settles seven Dita- λ points by certificate. The 512-point coarse grid already missed every known third-MUB locus (§VI-B); the same class of method produces these negatives, so they are confirmatory, not a substitute for Open Problem 1.

G. Reproducibility

Reproducibility is pinned in REPRODUCE.md (Julia 1.12.6, project-local .julia-depot). Search sets $\mathcal{S}_{588}$, $\mathcal{S}^*$, and $\mathcal{R}$ are formalized in docs/SEARCH_SETS.md; claims are tracked in results/final_honest_status.md (Proved / HP-verified / Conjectured / Open).

No fourth MUB was observed at any re-audited or certified-search point. Third MUBs exist at special Karlsson subfamilies identified above. At generic $K_6(0.3, 0.5, 0.2)$, max_clique = 2; the median pairwise inner product $|\langle v, w \rangle| \approx 0.353$ is an empirical graph statistic, not a proved obstruction. Planted-clique validation (Section VI-C) rules out clique-finder failure.

H. Comparison With Deprecated Approach

An earlier (incorrect) sweep used a *fixed* pool unbiased to $\{\mathbf{I}, F_6\}$, filtering only by additional unbiasedness to H . That pool had n_valid = 0 at all 1440 grid points—consistent with Grassl’s theorem [6]. The per- H reformulation is essential for testing Karlsson-family extendability.

VII. DISCUSSION

What this report claims. Finding 1 identifies two HP-verified third-MUB loci in the audited search. Findings 4, 2, 3 supply sampled persistence and boundary geometry. Open Problem 1 remains open.

Self-correction, and why it cuts both ways. An initial coarse 512-point grid incorrectly suggested max_clique = 2 everywhere; re-audit found third-MUB loci that the grid missed. That disclosure is methodological, but it is also a warning: the

same style of dense parameter sweep produced the *negative* fourth-MUB reports (CSV, $\mathcal{S}^*$, 628/628 Dita probes). A method that silently missed all known third-MUB loci is not, by itself, a reason to trust “no fourth MUB found” on a larger set. The certified seven-point obstruction [1] is the exception.

Explaining why the bounded λ -arc at F_6 and the full circle at Dita differ uses the companion gauge theorem [1]: at $\theta = 0$, ϕ is gauge; at Dita, λ is a genuine moduli direction. An analytic formula for κ in terms of Möbius data remains open (arc_circle_asymmetry.txt).

VIII. CONCLUSION

This computational report identifies two reproducing third-MUB loci in audited Karlsson search sets, documents a corrected per- H pipeline and its failure modes (including the 512-point grid miss), and records confirmatory fourth-MUB negatives on $\mathcal{S}^*$ and related sets without promoting them to theorems. Certified and exact fourth-MUB results, gauge algebra, and the A -matrix literature correction are in the companion theorems paper [1]. I do not claim a theorem on fourth-MUB absence in all of $K_6^{(3)}$ or on $N(6)$.

ACKNOWLEDGMENT

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APPENDIX A

LOCUS GEOMETRY AND DITA THIRD-MUB FINDINGS

Findings 2, 3, 4, 5 record HP computations.

APPENDIX B

LOCUS GEOMETRY

Define the *clique number* $\kappa(H)$ as the maximum clique size in the orthogonality graph on the deduplicated MU-pool of vectors unbiased to $\{\mathbf{I}, H\}$ (certified per- H solve).

Finding 2 (One-dimensional third-MUB locus at Dita (HP)). *Let $\theta_D = \arccos(1/\sqrt{3})$. On the probes below, $\kappa(H(\theta_D, \phi, \lambda)) \geq 6$ occurs at $\phi = \pi/4$ and not at $|\Delta\phi| \geq 10^{-3}$. For $\phi = \pi/4$, $\kappa \geq 6$ at each tested λ on the Dita circle (126/126 and 628/628). The all- λ inference uses 2π -periodicity from the companion gauge theorem (Paper 1) and is not a closed-form proof.*

a) *Computational support.*: Off $\phi = \pi/4$, HP axis probes show $\kappa \leq 2$ for $|\Delta\phi| \geq 10^{-3}$ at nine test λ (locus_phi_sweep_full_circle.jl). A 20×20 grid in (ϕ, λ) at fixed θ_D finds 0/400 clique-6 hits because grid points miss $\phi = \pi/4$ by $\geq 2.6 \times 10^{-3}$ —the same class of miss as the 512-point coarse grid. On $\phi = \pi/4$, full-circle λ persistence is an HP re-verification (lambda_periodicity_dita.txt).

Finding 3 (Bounded λ -arc at $\theta = 0$ (HP)). *At $\theta = 0$, ϕ is a parametrization gauge (Paper 1), and $\kappa(H(0, \phi, \lambda)) \geq 6$ only for λ in a bounded interval of width ≈ 0.118 rad about $\lambda_0 = 0.3$ (HP boundaries at $|\Delta\lambda| = 0.055044803035$ and 0.0625984193874 ; f6_boundary_reverify.txt).*

b) *Computational support.*: Gauge independence of ϕ at $\theta = 0$ (Paper 1) reduces ϕ -dependence. Binary search on λ at 400-bit precision locates clique-6 drop boundaries; interior points on the arc retain clique-6 by HP audit. This is a numerical boundary location, not an algebraic characterization of the arc.

APPENDIX C

THIRD MUB ON THE DITA λ -CIRCLE (FINDING T2)

Finding 4 (Third MUB on sampled Dita- λ). *Let $\theta_D = \arccos(1/\sqrt{3})$ and $\phi = \pi/4$. Write $H(\lambda) := H(\theta_D, \phi, \lambda)$. At each tested λ on the Dita circle (126/126 periodicity samples on $[0, 2\pi]$ and 628/628 dense probes), the pair $\{\mathbf{I}, H(\lambda)\}$ admits a third mutually unbiased basis obtained by certified pool extraction. By 2π -periodicity (Paper 1) this construction is 2π -periodic in λ . An informal continuity argument (Route B below) suggests the same for all $\lambda \in \mathbb{R}$; that extension is **not** a theorem.*

a) *Computational support.*: Fix λ . Solve the per- H MU-pool polynomial system (10 equations, 10 variables) at $H(\lambda)$ via certified homotopy continuation (generate_candidate_pool_fresh). Let $\mathcal{P}(\lambda)$ be the deduplicated pool of MU vectors. If the orthogonality graph on $\mathcal{P}(\lambda)$ contains a 6-clique C , the columns of $B_3 = [v_{c_1}, \dots, v_{c_6}]$ form a third ONB.

Anchor. At $\lambda_0 = 0.4$, Brierley and Weigert (2009) established third MUB existence at Dita-type parameters; the pipeline recovers `max_clique=6`.

Sampled persistence. Periodicity $H(\lambda + 2\pi) = H(\lambda)$ (Paper 1). High-precision verification (verify_dita_construction.jl, lambda_periodicity_dita.jl):

- 126/126 samples on $[0, 2\pi]$ with clique ≥ 6 ;

- 628/628 dense λ probes with `found_fourth=false` (numerical; the certified seven-point obstruction is in Paper 1);
- HP re-verification (256–400 bit) of clique orthogonality at spot checks.

CHM-inequivalence of distinct $H(\lambda)$ on the Dita slice (Paper 1) shows this is not a single-matrix artifact.

Informal extension (Route B). On intervals between λ -values where pool completeness holds, the clique template varies continuously when roots are certified. Combined with 2π -periodicity and the full-circle HP audit, this *suggests* third MUB existence for all $\lambda \in \mathbb{R}$. The continuity step is not an analytic proof; a closed-form $B_3(\lambda)$ remains open.

Finding 5 (Per- H pool zero-dimensionality at Dita, L7). *For $H = H(\theta_D, \pi/4, 0.4)$, HomotopyContinuation certifies 240 distinct isolated roots of the MU-pool system (mixed volume 252; `witness_decomposition.jl`, `pool_Dita_exact.m2`).*

APPENDIX D THIRD-MUB LOCUS PROBE TABLES

TABLE IV
THIRD-MUB LOCUS PROBES AFTER GAUGE RESOLUTION (2026-08-03). ISOTROPIC: 100 GAUSSIAN SAMPLES PER CELL; AXIS: BINARY SEARCH, 400-BIT HIGH-PRECISION (HP).

Anchor	Probe	Clique-6	Note
Dita	iso. $r=10^{-2}$	0/100	all clique-2
Dita	$\Delta\lambda$	126/126	full circle
Dita	20×20 grid	0/400	misses $\phi=\pi/4$
Dita	$\Delta\phi$	drop 10^{-3}	9/9 λ
Dita	$\Delta\theta$	drop 10^{-6}	
$F_6(\theta=0)$	iso. $r=10^{-2}$	0/100	
$F_6(\theta=0)$	$\Delta\phi$	gauge	H ϕ -free
$F_6(\theta=0)$	$\Delta\lambda$	arc	width ≈ 0.118
$F_6(\theta=0)$	$\Delta\theta$	drop 10^{-6}	

TABLE V
DITA ϕ -OFFSET SWEEP AT NINE λ VALUES (400-BIT HP, `LOCUS_PHI_SWEEP_FULL_CIRCLE.JL`).

λ	clique@ $\pi/4$	clique@ $\pi/4+0.001$	HP
0.3	6	2	yes
0.4	6	2	yes
0.5	6	2	yes
1.0	6	2	yes
2.0	6	2	yes
3.0	6	2	yes
4.0	6	2	yes
5.0	6	2	yes
6.0	6	2	yes

APPENDIX E REPRODUCTION AND SYMBOLIC LOGS

One-command regeneration of all artifacts is documented in `REPRODUCE.md`. Key deliverables: `results/special_loci_search.csv` (928 rows), `results/special_loci_degen1845.meta.txt` (S^* primary: 1863/1865, log-backed), `results/locus_classification.json`, `results/dita_lambda_fourth_dense.csv` (628 probes), `results/project_finish_audit.txt`, `results/track_c_elimination/m2_fourth_mub_reduced_Dita.log` (35 generators, $\dim I = -1$ over $\mathbb{C}$, inconclusive). The claim ledger (`results/final_honest_status.md`) tracks each statement as Proved / HP-verified / Conjectured / Open.

REFERENCES

- [1] A. Jaupi, “Gauge Structure and Certified Non-Extendability for Selected Karlsson Hadamards in Dimension Six,” companion theorems manuscript to the present methods report, 2026.
- [2] G. Zauner, “Quantendesigns—Grundzüge einer nichtkommutativen Designtheorie,” Ph.D. dissertation, Univ. Wien, 1999.
- [3] A. J. Scott and M. Grassl, “SIC-POVMs and MUBs from algebraic number theory,” *J. Math. Phys.*, vol. 51, no. 4, p. 042203, 2010.
- [4] I. Bengtsson, W. Tadej, and K. Życzkowski, “Five qubit entangled states, positive maps, and matrix pencils,” *Phys. Rev. A*, vol. 75, no. 3, p. 032319, 2007.
- [5] S. Brierley and S. Weigert, “Constructing mutually unbiased bases in dimension six,” *Phys. Rev. A*, vol. 79, no. 5, p. 052316, 2009. [Online]. Available: <https://arxiv.org/abs/0901.4051>
- [6] M. Grassl, “On SIC-POVMs and MUBs in dimension six,” arXiv:quant-ph/0405174, 2004.

- [7] A. Karlsson, “Three-parameter families of complex Hadamard matrices of order 6,” arXiv:1003.4177, 2011.
- [8] D. McNulty and S. Weigert, “Mutually Unbiased Bases in Composite Dimensions – A Review,” *Quantum*, vol. 10, p. 2051, 2026.
- [9] U. Haagerup, “Orthogonal maximal abelian $\ast$ -subalgebras of the 6×6 matrices,” in *Proc. Internat. Conf. Operator Algebras*, 2001.
- [10] G. Björck and R. Fröberg, “A faster way to count the solution set of trigonometric counting problems through matrix multiplication,” *Exp. Math.*, vol. 24, no. 4, pp. 468–473, 2015.
- [11] P. Breiding and S. Timme, “HomotopyContinuation.jl: A package for homotopy continuation in Julia,” in *Proc. ICMS*, 2018. [Software] Version 2.22.